

Brown University
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Eric Wu

education

Sc.B., Computer Science & A.B. Philosophy, *Brown University*, 2024–2028 (expected).

Coursework: *Logic for Systems, Distributed Computer Systems, Deep Learning, Computer Systems, Abstract Algebra, Statistics & Probability w/ Calculus, Theoretical Linear Algebra.*

research experience

Incoming Undergraduate Research Assistant, GIRAFFE Lab, *Brown University*, starting October 2026.

(Primary advisor: Prof. Serena Booth.)

- Investigating collusion in multi-agent reinforcement learning environments.

Research Fellow, *USC Information Sciences Institute*, April 2026–Present.[†]

(Primary advisor: Prof. Luca Luceri & Dr. Lanqin Yuan.)

- Built an end-to-end LLM social media simulation pipeline combining inverse reinforcement learning (IRL) with the OASIS multi-agent framework via vLLM across distributed compute.
- Improved realism of agent-based simulated social media behavior, reducing median KL divergence from empirical data by 73.5% via IRL-derived reward injection into agent personas.
- Evaluated fidelity using KL divergence, Wilcoxon signed-rank tests ($p < 0.05$), network topology metrics and NLP-driven semantic analysis (TweetNLP) of generated content.

[†] Fellowship and work supported by the National Science Foundation.

Undergraduate Research Assistant, Automatic Coordination of Teams Lab,
Brown University, March 2025–September 2025.

(Primary advisor: Tabitha Oanda, PhD Student.)

- Engineered new features into a multi-agent simulation environment, including dynamic obstacles and physics modifications to study coordination under realistic constraints.
- Designed and ran controlled experiments comparing heuristic algorithms, analyzing their effects on training stability, convergence, and behavioral variability.
- Deployed large-scale experiment batches on high-performance computing clusters with Slurm scheduling and used TensorBoard to analyze agent performance trends and policy generalization.

professional experience

Software Engineering Intern, Digital Investment Technologies,
Royal Bank of Canada, May 2024–August 2024.

- Orchestrated automatic financial plan verification via full-stack application (Spring Boot, Angular, Hadoop, Oracle), reducing manual review time by 37.5 hrs/week.
- Boosted plan reliability by 20% through validation against 34 business rules and 100+ automated tests to flag inconsistencies and errors.
- Led front-end development of a platform used across teams, aligning with RBC design guidelines and collaborating with designers to ensure accessibility and usability.

teaching experience

Undergraduate Teaching Assistant, *CSCI 0300: Fundamentals of Computer Systems*, Brown University, Spring 2026–Fall 2026 (present).

projects

Cellular Automata Formal Methods Model Checker.

([Github](#) • [Oral Presentation Slides](#))

- Co-developed a formal model of 1D cellular automata in Forge, verifying rule properties (injectivity, monotonicity, additivity) with automatic counterexample generation, and visualizer for generated traces.
- Implemented soundness and verification procedures for Garden of Eden configurations and strong twin states.

Ailurus: Webscraper & Update Notifier.

([Github](#))

- Built a web scraper to monitor a popular fiction platform and detect newly published/updated webnovels, triggering alerts via webhook on an hourly cron schedule.
- Enabled real-time user notifications via Discord API to ensure immediate awareness of new publications.
- Dockerized and deployed the app on DigitalOcean with GitHub Actions, enabling low-maintenance CI/CD.

skills

Languages: Java, Python, C, C++, SQL, JavaScript, TypeScript, Lua, $\text{\LaTeX}$.

Frameworks: Node.js, Angular, JUnit, Mockito, Spring, Spring Boot, Material-UI, Express.

Tooling: Git, DigitalOcean, Docker, GitHub Actions, Jupyter Notebooks, VS Code.

Libraries: PyTorch, Tensorflow, pandas, NumPy, Matplotlib, BeautifulSoup, Cheerio, vLLM, Ollama.

service to the community

Officer, Outreach & Partnerships, *Brown Votes*, September 2024–May 2026.

Intern, Philanthropy, *United Way of Rhode Island*, September 2025–May 2026.

Volunteer, *Student Clinic for Immigrant Justice*, January 2025–June 2025.

extra- curriculars

Executive Board, *Brown/RISD Game Developers*, 2024–Present.

Executive Board, *Brown Cooking Club*, 2024–Present.

Developer/Designer, *Full Stack at Brown*, 2024–Present.

Tour Guide, *Office of College Admissions, Brown University*, 2025–Present.

Editor, *A Priori: Brown Undergraduate Journal of Philosophy*, 2024–Present.